

2019 Mathematics Kangaroo Contest

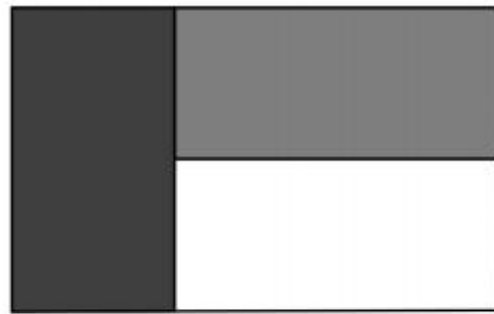
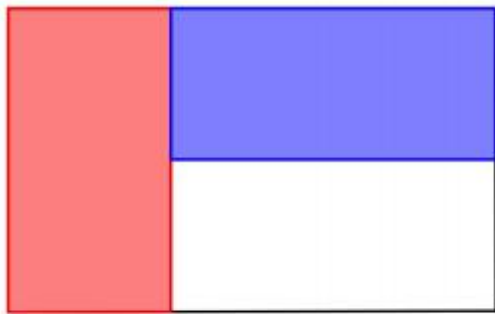
—Cadet—

2019 数学袋鼠竞赛

Part 1: 10 problems, 3 points each | 第一部分: 10 道题目, 每题 3 分

1. The flag of Kangoroland is a rectangle which is divided into three smaller equal rectangles as shown. What is the ratio of the side lengths of the white rectangle?

袋鼠国的国旗是一个长方形，被分成了三个同样大小的长方形。问白色长方形的两条边的比例是多少？



(A) 1 : 2

(B) 2 : 3

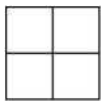
(C) 2 : 5

(D) 3 : 7

(E) 4 : 9

2. The numbers 1, 2, 3 and 4 are each written in different cells of the 2×2 table. After that, the sum of the numbers in each row and column is calculated. Two of these sums are 4 and 5. What are the other two sums?

将数字 1, 2, 3, 4 填入 2×2 方格表的 4 个不同方格中，并计算各行与各列的总和。其中，两个总和是 4 和 5，问另两个总和是多少？



(A) 6, 6

(B) 3, 5

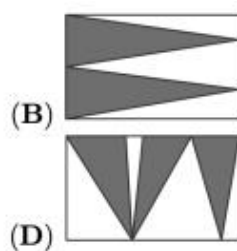
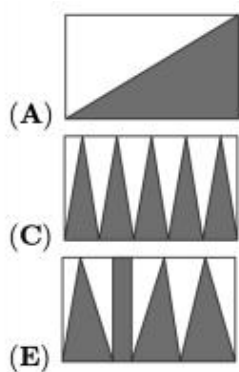
(C) 4, 5

(D) 4, 6

(E) 5, 6

3. A rectangle has been shaded in five different ways as shown. In which diagram does the shaded part have the largest area?

一个长方形以如下五种不同的方式染色。问哪一个图形中的阴影部分的面积最大？

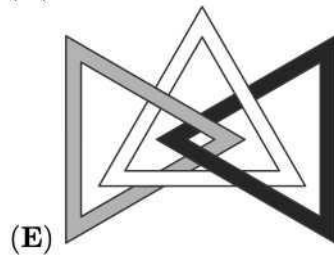
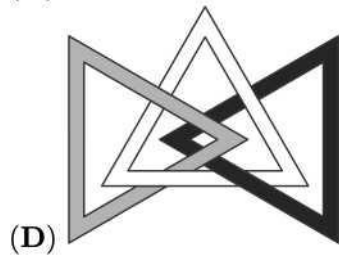
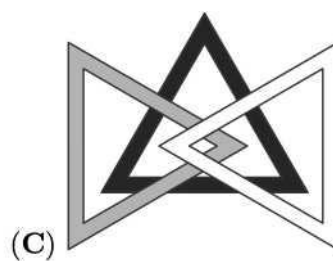
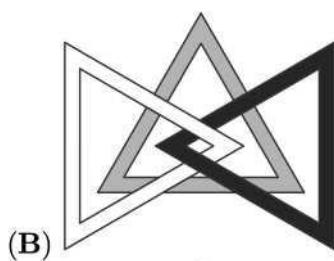
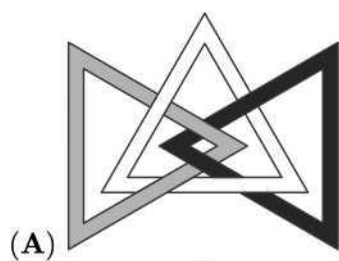


4.



Three triangles are linked as shown above. Which of the following pictures shows these three triangles linked in the same way?

如上所示，三个三角形相互缠绕。问下面哪个图中的三个三角形有相同的缠绕方式？



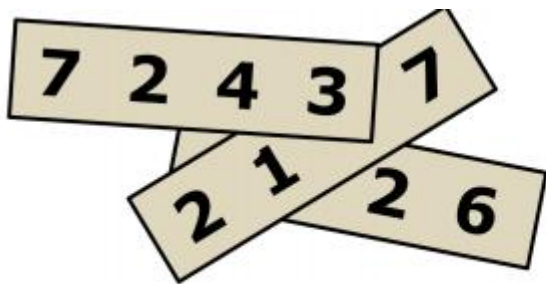
5. A pyramid has 23 triangular faces. How many edges does this pyramid have?

如果一个棱锥有 23 个三角形的面，问这个棱锥有多少条边？

- (A) 23 (B) 24 (C) 46 (D) 48 (E) 69

6. Three 4-digit numbers are written on three pieces of paper as shown. The sum of the three numbers is 11126. Three of the digits are covered. What are the covered digits?

三张纸片上分别写有三个四位数，并且这三个数的和是 11126。有三个数字被遮盖住了，问被遮盖住的数字是什么？



- (A) 1, 4, 7 (B) 1, 5, 7 (C) 3, 3, 3 (D) 4, 5, 6 (E) 4, 5, 7

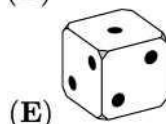
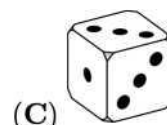
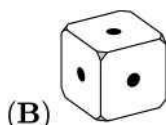
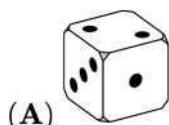
7. What is the first (leftmost) digit of the smallest positive integer whose digits add up to 2019?

最小的各位数字之和等于 2019 的正整数的首位(最左边)数字是几？

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

8 • Each of the faces of a die is marked with either 1, 2 or 3 dots so that the probability of rolling a 1 is $\frac{1}{2}$, the probability of rolling a 2 is $\frac{1}{3}$, and the probability of rolling a 3 is $\frac{1}{6}$. Which of the following cannot be a view of this die?

一个骰子的每个面上标有 1 个，2 个，或者 3 个点，并且它掷出 1 的概率是 $\frac{1}{2}$ ，掷出 2 的概率是 $\frac{1}{3}$ ，掷出 3 的概率是 $\frac{1}{6}$ 。问下面哪个不可能是这个骰子的视图？



9 • Michael invented a new $\diamond$ operation for real numbers, defined as $x \diamond y = y - x$. If a , b and c satisfy $(a \diamond b) \diamond c = a \diamond (b \diamond c)$, which of the following statements is necessarily true?

Michael 定义了一个关于实数的新运算 $\diamond$ ，为 $x \diamond y = y - x$ 。如果 a , b 和 c 满足 $(a \diamond b) \diamond c = a \diamond (b \diamond c)$ ，问下面哪个论断必然是正确的？

- (A) $a = b$ (B) $b = c$ (C) $a = c$ (D) $a = 0$ (E) $c = 0$

10 • How many of the numbers from 2^{10} to 2^{13} , inclusive, are divisible by 2^{10} ?

在 2^{10} 到 2^{13} 之间(包括首尾二数), 共有多少个能被 2^{10} 整除?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 16

Part 2: 10 problems, 4 points each | 第二部分: 10 道题目, 每题 4 分

11. Which is the highest power of 3 dividing the number $7! + 8! + 9!$?

能整除数 $7! + 8! + 9!$ 的 3 的最高方幂是多少?

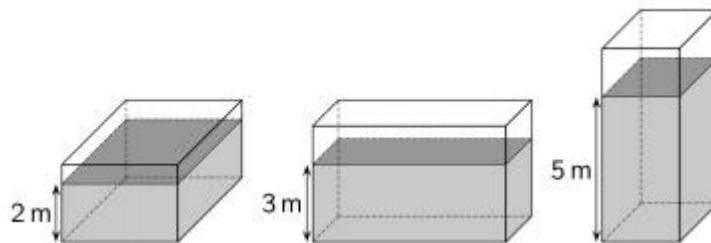
- (A) 3^2 (B) 3^4 (C) 3^5 (D) 3^6
 (E) a power of 3 higher than 3^6 一个比 3^6 还高的 3 的方幂

12 • This year, the number of boys in my class has increased by 20% and the number of girls has decreased by 20%. We now have one student more than before. Which of the following could be the number of students in my class now?

今年, 我的班级中男生的人数增加了 20%, 而女生的人数减少了 20%。现在比以前多一个学生。问下列哪个数可能是现在班级里的学生数?

- (A) 22 (B) 26 (C) 29 (D) 31 (E) 34

13.



A container in the shape of a rectangular box is partially filled with 120 cubic meters of water. The depth of the water is either 2 meters or 3 meters or 5 meters, depending on which side of the box is on the ground, as shown above (not to scale). What is the volume of the container in cubic meters?

一个长方体容器中装了 120 立方米的水, 但没有装满。如上图所示(未按比例绘制), 取决于哪个面放在地上, 不同摆放方式下的水深分别为 2 米、3 米或 5 米。问此容器的体积是多少立方米?

- (A) 160 (B) 180 (C) 200 (D) 220 (E) 240

14. Three kangaroos, Alex, Bob and Carl, go for a walk every day. If Alex doesn't wear a hat, then Bob wears a hat. If Bob doesn't wear a hat, then Carl wears a hat. Today Carl is not wearing a hat. Who is certainly wearing a hat today?

三只袋鼠 Alex, Bob 和 Carl 每天一起散步。如果 Alex 不戴帽子, 那么 Bob 会戴帽子。如果 Bob 不戴帽子, 那么

Carl 会戴帽子。今天 Carl 没戴帽子。问谁今天一定戴帽子了？

(A) only Alex and Bob 只有 Alex 和 Bob

(B) only Alex 只有 Alex

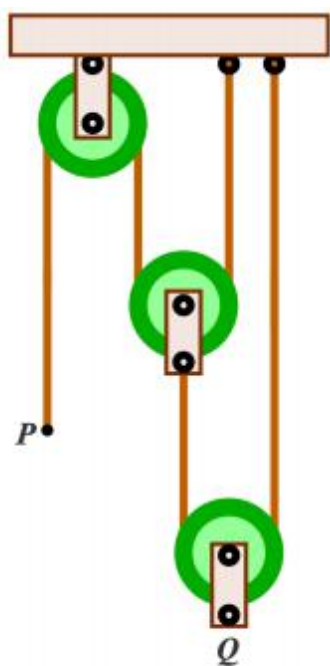
(C) Alex, Bob and Carl Alex、Bob 和 Carl

(D) neither Alex nor Bob 既不是 Alex 也不是 Bob

(E) only Bob 只有 Bob

15. The system shown consists of three pulleys with vertical sections of rope between them. The end P is moved down 24 centimeters. How many centimeters does point Q move up?

下面的滑轮系统是由三个滑轮和他们之间竖直的绳子组成。如果 P 端被拉下 24 厘米，问 Q 点会升高多少厘米？



(A) 24

(B) 12

(C) 8

(D) 6

(E) $\frac{24}{5}$

16 • A positive integer n is called good if its largest divisor (excluding n) is equal to $n-6$. How many good positive integers are there?

一个正整数 n 被称为“好数”，如果它的最大的因子(不包括 n 本身)等于 $n-6$ 。问一共有多少个正整数是“好数”？

(A) 1

(B) 2

(C) 3

(D) 6

(E) infinitely many 多

17 • A box contains 4 chocolates and 1 fruit chew. John and Mary take turns drawing a treat out of the box without replacement. Whoever draws the fruit chew wins. John draws first. What is the probability that Mary wins?

一个盒子里放着 4 包朱古力糖和 1 包水果糖。John 和 Mary 轮流从盒子里抽出一包糖，抽出的糖不再放回，谁先抽到水果糖谁就算赢。如果 John 先抽的话，Mary 获胜的概率是多少？

(A) $\frac{2}{5}$

(B) $\frac{3}{5}$

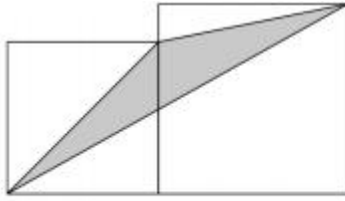
(C) $\frac{1}{2}$

(D) $\frac{5}{6}$

(E) $\frac{1}{3}$

- 18 • Two adjacent squares with side lengths a and b ($a < b$) are shown. What is the area of the shaded triangle?

如图所示，两个相邻的正方形边长分别是 a 和 b ($a < b$)。问阴影三角形的面积是多少？



- (A) $\sqrt{ab}$ (B) $\frac{1}{2}a^2$ (C) $\frac{1}{2}b^2$ (D) $\frac{1}{4}(a^2 + b^2)$ (E) $\frac{1}{2}(a^2 + b^2)$

19. What is the integer part of $\sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20}}}}}$?

$\sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20}}}}}$ 的整数部分是多少？

- (A) 4 (B) 5 (C) 6 (D) 20 (E) 25

20. To calculate the result of $\frac{a+b}{c}$, Sara types $a + b \div c =$ on a calculator and the result is 11 (a , b , and c are positive integers). She then types $b + a \div c =$ and she is surprised to see that the result is 14. She realizes that the calculator is designed to calculate divisions before additions. What is the correct result of $\frac{a+b}{c}$?

为了计算 $\frac{a+b}{c}$ 的结果，Sara 在计算器上输入 $a + b \div c =$ ，并得到结果是 11 (a ， b 和 c 是正整数)。她接着又输入了 $b + a \div c =$ ，并惊讶的发现结果是 14。她随即想起计算器是按照“先乘除，后加减”的顺序计算。问 $\frac{a+b}{c}$ 的正确结果应该是多少？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Part 3: 10 problems, 5 points each | 第三部分：10 道题目，每题 5 分

21. Let a be the sum of all positive divisors of 1024 and b the product of all positive divisors of 1024. Then 设 a 是 1024 的所有正因子的总和， b 是 1024 的所有正因子的乘积。那么有？

- (A) $(a-1)^5 = b$ (B) $(a+1)^5 = b$ (C) $a^5 = b$ (D) $a^5 - 1 = b$ (E) $a^5 + 1 = b$

- 22 • What is the set of all values of the parameter a for which the number of solutions of the equation

$2 - |x| = ax$ is equal to two?

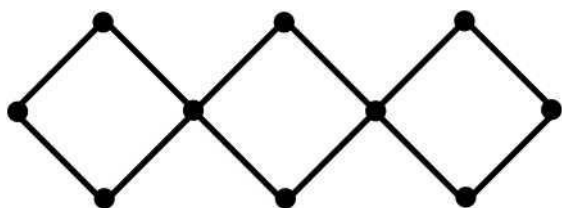
问使得方程 $2 - |x| = ax$ 的解的个数恰为两个的所有参数 a 的集合是什么？

- (A) $(-\infty, -1]$ (B) $(-1, 1)$ (C) $[1, +\infty)$ (D) $\{0\}$ (E) $\{-1, 1\}$

- 23 • The vertices of the network shown are labelled with the numbers from 1 to 10. The sum S of the four labels

on each square is the same. What is the least possible value of S ?

下图中网络的各项点处分别标有从 1 到 10 的数。假设每个正方形的四个顶点处所标数的总和都等于 S 。问 S 的最小可能值是多少?



- (A) 18 (B) 19 (C) 20 (D) 21 (E) 22

24 • How many planes pass through at least three vertices of a given cube?

问共有多少个平面包含一个给定立方体的至少三个顶点?

- (A) 6 (B) 8 (C) 12 (D) 16 (E) 20

25. Four distinct straight lines pass through the origin of the coordinate system. They intersect the parabola $y = x^2 - 2$ at eight points. What can be the product of the x-coordinates of these eight points?

有四条过原点的不同直线。它们和抛物线 $y = x^2 - 2$ 交于 8 个点。问这 8 个点的 x 坐标的乘积可能是多少?

- (A) only 16 只能是 16 (B) only -16 只能是 -16
 (C) only 8 只能是 8 (D) only -8 只能是 -8
 (D) There are several possible products. 有很多种可能的乘积

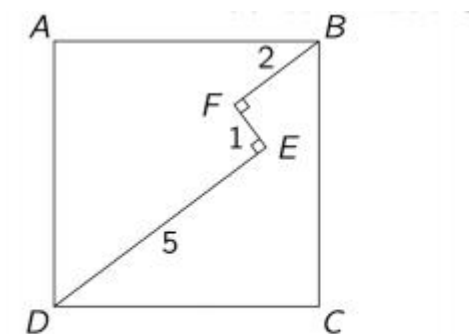
26. For how many integers n is $|n^2 - 2n - 3|$ a prime number?

有多少个整数 n , 使得 $|n^2 - 2n - 3|$ 是素数?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) infinitely many 无穷多

27 • A path $DEFB$ with $DE \perp EF$ and $EF \perp FB$ lies inside the square $ABCD$ as shown. Given that $DE = 5$, $EF = 1$ and $FB = 2$, what is the length of the side of the square?

如图所示，正方形 $ABCD$ 中有一条折线 $DEFB$ 满足 $DE \perp EF$, $EF \perp FB$ 。已知 $DE = 5$, $EF = 1$ ，并且 $FB = 2$ ，问正方形的边长是多少？



(A) $3\sqrt{2}$

(B) $\frac{7\sqrt{2}}{2}$

(C) $\frac{11}{2}$

(D) $5\sqrt{2}$

(E) none of the previous 不是以上的数

28 • The sequence $a_1, a_2, a_3, \dots$ starts with $a_1 = 49$. For $n \geq 1$, the number a_{n+1} is obtained by adding 1 to the sum of the digits of a_n and then squaring the result. Thus $a_2 = (4 + 9 + 1)^2 = 196$. Determine a_{2019} .

数列 $a_1, a_2, a_3, \dots$ 起始于 $a_1 = 49$ 。当 $n \geq 1$ 时， a_{n+1} 等于 a_n 的各位数字之和加 1，然后再平方得到的结果。例如， $a_2 = (4 + 9 + 1)^2 = 196$ 。求 a_{2019} 。

(A) 121

(B) 25

(C) 64

(D) 400

(E) 49

29 • Three different numbers are chosen at random from the set $\{1, 2, 3, \dots, 10\}$. What is the probability that one of them is the average of the other two?

从集合 $\{1, 2, 3, \dots, 10\}$ 中随机选取三个不同的数。其中的一个数恰好等于另外两数的平均数的概率是多少？

(A) $\frac{1}{10}$

(B) $\frac{1}{6}$

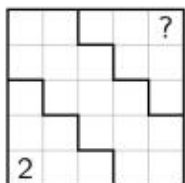
(C) $\frac{1}{4}$

(D) $\frac{1}{3}$

(E) $\frac{1}{2}$

30 • The square shown is filled with numbers in such a way that each row and each column contains the numbers 1, 2, 3, 4 and 5 exactly once. Moreover, the sum of the numbers in each of the three bold-bordered areas is equal. What number is in the upper right corner?

下图的方格表的每行和每列中，数 1, 2, 3, 4, 5 恰好各出现一次。并且图中的三块以粗线为边界的区域中的所有数的总和都相等。问右上角的那个数是几？



(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

答案: AEEDC BBCDD DBEED CABAE BBCEA DECBC